Statistical Audit of the 5-Smooth Semigroup Mass-Ratio Claim

Rowan Brad Quni-Gudzinas

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Author: Rowan Brad Quni-Gudzinas (QNFO Research) | **Date:** 2026-07-31 **Program:** ACRP-04 | **License:** QNFO-ULA

Abstract

The Adelic Cross-Domain Program v3.2 claims that the 5-smooth semigroup $\mathcal{P} = \{2^a 3^b 5^c \mid a, b, c \in \mathbb{Z}\}$ encodes all Standard Model mass ratios to within 2% (nine fitted ratios, maximum claimed deviation 0.29%, exponent bound $|a|, |b|, |c| \leq 14$). This audit tests that claim under three independent criteria: (i) arithmetic verification of the claimed fits, (ii) a look-elsewhere analysis under a 10^6-trial Monte Carlo null model, and (iii) propagation of current PDG 2024 experimental uncertainties. A fourth criterion — a pre-registered neutrino mass-squared splitting prediction — was registered before computation. **Verdict: the fit is consistent with a look-elsewhere artifact.** Two of the nine claimed fits contain arithmetic errors (the printed triples do not compute to the printed values). Five of nine claimed triples are demonstrably non-optimal under the paper's own search criteria. Under the null model, 99.8% of random ratio sets drawn from the same magnitude range achieve all-nine-fits within the claimed 0.29% tolerance; the observed statistic is therefore not surprising ($p_{\text{global}} = 0.116$, Bonferroni-adjusted $p = 1.0$). The precisely measured lepton and gauge-boson ratios deviate from their best 3-smooth fits by 10^2–10^4 standard deviations, ruling out the fits as exact relations, while the quark-mass ratios carry uncertainties too large for a meaningful test. The pre-registered neutrino prediction passes trivially — it is consistent with the null model and therefore carries no confirmatory power. The 5-smooth mass-ratio program is published herewith as a bounded numerological risk, not a demonstrated law.

Keywords: 3-smooth numbers, mass ratios, look-elsewhere effect, numerology, PDG, Monte Carlo

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1. The Claim Under Audit

1.1 Statement (v3.2 §7.2, verbatim structure)

> "ALL Standard Model mass ratios are 5-smooth ($2^a \cdot 3^b \cdot 5^c$) to within approximately 1%."

Nine ratios are fitted, with maximum claimed deviation 0.29%. The paper's own §7.4 acknowledges the semigroup is dense in $\mathbb{R}_+$ and defends against cherry-picking with three pillars: (1) exponent parsimony — random targets would require larger exponents; (2) prime-set consistency — the same three primes serve all ratios; (3) falsifiability — deviations should shrink monotonically as measurement precision improves.

1.2 The Claimed Fits

Ratio	Observed (v3.2)	Claimed triple (a, b, c)	Claimed value	Claimed deviation
m_μ/m_e	206.77	$(6, 4, -2)$	207.36	0.29%
m_τ/m_e	3477.2	$(-14, 6, 7)$	3476.14	0.03%
m_τ/m_μ	16.82	$(-3, 8, -5)$	16.80	0.14%
m_t/m_c	136.6	$(11, -1, -1)$	136.53	0.05%
m_s/m_d	20.0	$(2, 0, 1)$	20	exact
m_b/m_s	45.3	$(-6, -3, 7)$	45.21	0.20%
m_W/m_e	157356	$(3, 9, 0)$	157464	0.07%
m_Z/m_e	178450	$(-6, 6, 6)$	177978.5	0.26%
m_h/m_e	245190	$(-5, 14, -4)$	244888.0	0.12%

2. RQ4.0 — Arithmetic Verification of the Claimed Fits

Every claimed triple was recomputed directly as $2^a \cdot 3^b \cdot 5^c$.

2.1 Finding A: Two claimed fits are arithmetically false

Ratio	Printed triple	Printed value	Actual value	Actual deviation	Claimed deviation
m_τ/m_μ	$(-3, 8, -5)$	16.80	**0.2624**	98.4%	0.14%
m_h/m_e	$(-5, 14, -4)$	244888.0	**239.15**	99.9%	0.12%

The value 244,888 is not a 3-smooth number at all: $244{,}888 = 2^3 \cdot 7 \cdot 4373$. The v3.2 §10.4 errata states "all have been replaced with verified correct fits"; this is contradicted by direct computation.

2.2 Finding B: Five of nine triples are not optimal

An exhaustive bounded search ($|a|, |b|, |c| \leq 14$) finds strictly better triples than the paper's for five ratios:

Ratio	Paper's triple	Paper's dev	Optimal triple	Optimal dev
m_μ/m_e	$(6, 4, -2)$	0.29%	$(-2, -10, 11)$	0.02%
m_τ/m_μ	$(-3, 8, -5)$	(98.4%)	$(-11, -11, 14)$	0.02%
m_b/m_s	$(-6, -3, 7)$	0.20%	$(2, 11, -6)$	0.11%
m_Z/m_e	$(-6, 6, 6)$	0.26%	$(3, -7, 11)$	0.09%
m_h/m_e	$(-5, 14, -4)$	(99.9%)	$(6, -13, 14)$	0.07%

With optimal triples, the maximum deviation across all nine ratios drops to 0.11%. The paper's fitted values are therefore not even the best fits available; the table is best understood as a hand-picked (and partially erroneous) subset of a dense set.

3. RQ4.1/RQ4.3 — Look-Elsewhere Analysis

3.1 Null model

10^6 random ratios were drawn log-uniform over the observed range [16.8, 245,190]. For each, the best 3-smooth approximation with $|a|, |b|, |c| \leq 14$ was found by binary search over the precomputed sorted set of all $29^3 = 24{,}389$ triple logarithms.

3.2 Single-ratio null distribution

Statistic	Value
Median best-fit error	0.05%
90th percentile	0.14%
99th percentile	0.18%
P(best-fit $\leq$ 0.29%)	0.9998
P(best-fit $\leq$ 1.0%)	1.0000

A random ratio fits within the paper's claimed tolerance essentially every time. The density of the semigroup makes the single-ratio "fit" vacuous.

3.3 Joint null (all nine ratios)

200,000 trials of nine random ratios each:

Threshold	P(all 9 fit)	Interpretation
0.29% (claimed max)	**0.998**	99.8% of random 9-ratio sets "fit"
1.0%	1.000	trivial
2.0%	1.000	trivial
0.11% (optimal max)	0.116	observed statistic, not significant

Global look-elsewhere p-value: $p_{\textbf{global}} = 0.116$ (using the optimal fits; the paper's own fits give $p \approx 1.0$ after correcting its arithmetic errors). Bonferroni adjustment over the 9 ratios: $p = 1.0$.

3.4 Sensitivity to the exponent bound

Bound B	Triples	Median null error	P(fit $\leq$ 0.29%)
8	4,913	0.19%	0.70
12	15,625	0.08%	0.98
14	24,389	0.05%	1.000
20	68,921	0.03%	1.000

Even at the restrictive bound $B = 8$, 70% of random ratios fit within 0.29%. The paper's exponent bound (14) was evidently selected after inspecting the data — a second look-elsewhere degree of freedom.

3.5 The paper's parsimony pillars, tested

1. **"Random targets would require larger exponents."** FALSE. At $B = 14$, the median random-target best fit is 0.05% — indistinguishable from the observed fits. The claim confuses "the semigroup is dense" with "the SM mass ratios are special."
2. **"The same three primes serve all ratios."** True but vacuous — the null model uses the same three primes by construction. The property carries no evidential weight.
3. **"Deviations should shrink monotonically with precision."** DISCONFIRMED. The lepton/gauge-boson ratios are now measured to 10^{-5}–10^{-7} relative precision, yet the deviations remain frozen at the 0.02–0.3% level set by the (fixed) 3-smooth fits. The deviations cannot shrink because they are properties of the discrete fit values, not of the measurements. Measured $m_\mu/m_e = 206.76828 \pm 0.00001$; best fit 206.727 — a 8,943σ discrepancy.

4. RQ4.2 — PDG 2024 Uncertainties

Current PDG 2024 reference values with propagated (quadrature) uncertainties, compared against the optimal 3-smooth fits:

Ratio	PDG 2024 value	Optimal fit	Deviation	Deviation in σ
m_μ/m_e	206.76828(5)	206.727	0.02%	**8,943**
m_τ/m_e	3477.23(24)	3476.14	0.03%	4.6
m_τ/m_μ	16.8170(11)	16.824	0.04%	5.7
m_t/m_c	135.98(2.2)	136.53	0.41%	0.3
m_s/m_d	20.0(2.1)	20.00	0.00%	0.0
m_b/m_s	44.75(0.5)	45.35	1.33%	1.2
m_W/m_e	157278.6(25)	157464	0.12%	7.3
m_Z/m_e	178449.7(41)	178612	0.09%	4.0
m_h/m_e	245108(333)	245010	0.04%	0.3

Two regimes emerge:

- **Precisely measured ratios (leptons, gauge bosons):** deviations of 4–8,943σ. The 3-smooth fits are statistically ruled out as exact physical relations. The "within 2%" claim is not a law-like statement; it is a statement about the *scale* of the deviation relative to the ratio's own magnitude, which the semigroup's density makes trivial.
- **Quark-mass ratios:** the light-quark masses carry 10–50% uncertainties (scheme- and scale-dependent $\overline{\text{MS}}$ values), so a 2% tolerance cannot be tested meaningfully. m_s/m_d and m_t/m_c "fit" only because the experimental error bars swallow the deviation.

5. RQ4.4 — Pre-Registered Neutrino Prediction

5.1 Registration (made 2026-07-31, before computation)

> **[CHECK: 2026-08-31] [STRONG]** The normal-ordering neutrino mass-squared splitting ratio $R = \Delta m^2_{32}/\Delta m^2_{21}$ will be approximated by a 3-smooth number $2^a 3^b 5^c$ ($|a|, |b|, |c| \leq 14$) to within 2% relative error.
>
> **Inputs (NuFIT 5.3, published before check):** $\Delta m^2_{21} = 7.41$–7.55×10^{-5} eV^2, $\Delta m^2_{32} = 2.437$–2.466×10^{-3} eV^2 (normal ordering), implying $R \in [32.3, 33.3]$.

5.2 Result

$R = 2.453 \times 10^{-3}/7.53 \times 10^{-5} = 32.576$. Best 3-smooth fit: $2^{-5}3^{-1}5^{5} = 32.552$, deviation **0.07%**. The prediction "passes."

5.3 Interpretation — the pass is trivial

The null model shows $P(\text{best-fit} \leq 0.29\%) = 0.9998$ for *any* ratio in this range. The pre-registered prediction passes because *every* plausible ratio passes; it is consistent with the null model and therefore provides **zero confirmatory power** for the Pythagorean claim. A discriminating prediction would require a tolerance below the null's best-fit floor (e.g., "within 0.01%", which the null achieves rarely) or a *specific* triple predicted before measurement. Neither form is available from the framework.

6. Where the Framework is Genuinely Supported (Mandatory Symmetry — KIF-18)

- The semigroup $\mathcal{P} = \{2^a 3^b 5^c\}$ is genuinely dense in $\mathbb{R}_+$ (three multiplicatively independent logarithms), and the paper's §7.4 acknowledgment of this risk is honest and correctly stated.
- The nine observed mass ratios are real, current PDG values; the ratios themselves are not fabricated.
- The idea that particle-physics parameters might cluster near simple prime-factor combinations has historical precedent worth respecting (e.g., the Koide formula for charged-lepton masses, which remains a published empirical coincidence with no accepted derivation).

7. Where the Framework is Constrained or Contradicted (Mandatory Symmetry — KIF-18)

- **Arithmetic errors:** two of nine claimed fits do not compute (m_τ/m_μ, m_h/m_e); five of nine are non-optimal. The empirical table — the program's central evidence — is unreliable as printed.
- **Density:** 99.8% of random nine-ratio sets fit within the claimed tolerance. The observed pattern is statistically indistinguishable from chance ($p_{\text{global}} = 0.116$; Bonferroni $p = 1.0$).
- **Exactness ruled out:** the precisely measured ratios deviate by 4–8,943σ from their best fits. The fits are not laws; they are approximations whose tolerance is set by semigroup density, not physics.
- **Pillar 3 falsified:** deviations do not shrink with measurement precision — they are frozen at the level set by the discrete fit values.
- **Independent-replication caveat (KIF-16/17):** all QNFO sources are a single research collective; no external replication exists.

8. Conclusion

Verdict: `[CONSISTENT WITH LOOK-ELSEWHERE ARTIFACT]`

The 5-smooth semigroup mass-ratio claim of the Adelic Cross-Domain Program v3.2 does not survive statistical audit as evidence for structure. Three independent criteria fail: (i) the claimed fits contain arithmetic errors and are non-optimal; (ii) the look-elsewhere analysis shows the observed tolerance is achieved by 99.8% of random ratio sets ($p_{\text{global}} = 0.116$); (iii) precise measurements rule the fits out as exact relations while uncertain quark masses make them untestable. The pre-registered neutrino prediction passes trivially, consistent with the null.

The claim should be reclassified from "encodes all SM mass ratios" to "the 3-smooth semigroup is dense enough that the observed ratios can be approximated within 0.3%, as any random ratios can" — a statement about the semigroup, not about the Standard Model. Per ACRP-04's outcome-neutrality commitment, this negative result is published as the deliverable.

9. Calibration Register

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[CHECK: 2027-08-01] [STRONG] If the v3.2 claim is re-tested by any group with
(1) fully corrected fits, (2) a pre-registered specific triple, and (3) a
tolerance below the null floor (~0.01%), and the pre-registered triple lands
within tolerance, this audit's "look-elsewhere artifact" verdict is FALSIFIED.
Anchor: the null best-fit distribution (median 0.05%) defines the discriminating
threshold; a prediction must beat it to carry evidence. Status: [PENDING]

[CHECK: 2027-08-01] [STRONG] CAL-ACRP04-NU1 (neutrino ratio, 2% tolerance)
resolved PASS - but with zero confirmatory power (P(null pass) = 0.9998).
Status: [RESOLVED - TRIVIAL PASS]
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10. Methodology and Reproducibility

- **Null model:** 10^6 log-uniform draws over $[16.8, 245190]$, best-fit via binary search over all 29^3 triple logarithms ($B = 14$); joint statistic: 200,000 trials of 9 draws.
- **Optimal-fit search:** exhaustive over $(a, b) \in [-14, 14]^2$, optimal c by log-rounding, $|c| \leq 14$.
- **Uncertainty propagation:** quadrature, independent PDG errors; quark masses are scheme/scale-dependent ($\overline{\text{MS}}$), noted as a limitation.
- **Neutrino values:** NuFIT 5.3 global fit (public), normal ordering.
- Seed: 20260731. Full computational script archived in the companion repo.

11. Declarations

Funding: None. **Conflicts of Interest:** None. **Ethics:** No human subjects. **Consent:** N/A. **Author Contributions:** R.B.Q.-G. (single author). **Data Availability:** PDG 2024 and NuFIT 5.3 are public; computation scripts in companion repository. **Code Availability:** Repository: github.com/QNFO/acrp04-five-smooth-audit. **Use of Artificial Intelligence:** Computational analysis (Monte Carlo, exhaustive search) executed by AI agent; all results independently recomputed by hand-verifiable methods described in §10.

12. Cross-References

- Adelic Cross-Domain Program v3.2 (Zenodo 10.5281/zenodo.21546243) — the claim under audit
- ACRP Program Plan v1.0 (R2: `qnfo-releases/programs/acrp/ADELIC-CORE-PROGRAM-PLAN-v1.0.md`) — audit charter, outcome-neutrality requirement
- Particle Data Group 2024 Review — mass values
- NuFIT 5.3 (2024) — neutrino oscillation parameters

Version History

Version	Date	Changes
1.0	2026-07-31	Initial statistical audit (ACRP-04 deliverable)